AAE590AI Individual Project - Free Edge Stresses

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Introduction

The study of free-edge interlaminar stresses is an important area in the mechanics of composite laminates. Due to the anisotropic nature of fiber-reinforced composite materials, individual plies within a laminate exhibit different mechanical responses depending on their fiber orientations. When plies with different orientations are bonded together and subjected to external loading, their differences in stiffness, Poisson's ratios, and extension-shear coupling result in internal forces that must be balanced to maintain compatibility and equilibrium. Near the free edges of the laminate, these interactions produce localized interlaminar stress concentrations, particularly interlaminar shear stress τ_{xz} and normal or peeling stress σ_z. These stresses are especially important because they can initiate delamination between adjacent plies, potentially leading to progressive damage and structural failure.

Classical Laminate Theory (CLT), a foundational mathematical model used to analyze laminated composite structures, is based on a two-dimensional plane-stress assumption in which out-of-plane stresses are neglected. Although this assumption is generally suitable for predicting the global in-plane mechanical response of thin laminates, it does not capture the three-dimensional stress concentrations that develop near free edges. In particular, the mismatch in transverse contraction between plies produces interlaminar peeling stresses, while differences in the in-plane shear response of angle plies produce interlaminar shear stresses. These stresses arise from the requirement to satisfy three-dimensional equilibrium and traction-free boundary conditions, despite being absent from conventional CLT predictions. Consequently, understanding the physical mechanisms responsible for these stresses is essential for evaluating the potential for delamination initiation in laminated composite structures.

Analytical determination of free-edge interlaminar stresses is challenging because the problem fundamentally involves three-dimensional elasticity, requiring the solution of coupled partial differential equations with appropriate boundary conditions. Furthermore, the stress fields near material interfaces and free edges can exhibit highly localized gradients and singular behavior, making exact analytical solutions difficult to obtain. However, approximate solutions can be developed by simplifying the three-dimensional stress distributions into equivalent force and moment resultants. By assuming characteristic boundary-layer dimensions and simplified triangular stress distributions, the interlaminar stresses can be estimated using free-body diagrams and equilibrium relationships. Although these approximations do not represent the exact three-dimensional stress field, they provide a physically meaningful method of understanding and estimating the interlaminar forces acting at individual ply interfaces.

The objective of this project is to investigate the ability of artificial intelligence (AI), specifically large language models (LLMs), to understand, derive, and calculate approximate free-edge interlaminar stresses in composite laminates. While modern AI models are capable of solving many conventional engineering problems, their ability to independently interpret complex three-dimensional force interactions and develop appropriate analytical simplifications remains uncertain. Preliminary testing demonstrated that an AI model could identify the governing three-dimensional equilibrium equations but struggled to translate these relationships into physically meaningful approximations and accurate numerical solutions. This raises the question of whether providing structured, physics-informed knowledge can improve the model's ability to reason through a mechanics problem that is not readily solved through conventional prompting.

To investigate this, a series of AI skills containing progressively different levels of technical guidance are developed and evaluated. The first skill, V1, focuses on the underlying physical mechanisms, boundary conditions, and force and moment balances, requiring the AI to independently formulate its own approximate analytical solution. The second skill, V2, provides the mathematical relationships and analytical derivations used to calculate the interlaminar stresses, allowing the AI to apply an established approximation without the same level of physical explanation. Finally, the two skills are combined to determine whether integrating physical reasoning with established mathematical formulations improves the accuracy and consistency of the AI-generated solutions. The resulting predictions are compared against reference calculations for several symmetric laminate stacking sequences, with particular attention given to errors in physical assumptions, analytical derivations, and numerical implementation.

Through this investigation, the project aims to evaluate not only the numerical accuracy of AI-generated engineering solutions but also the extent to which structured physics-informed instructions improve the model's ability to formulate and explain the underlying mechanics. By identifying the differences between incorrect physical modeling, inconsistent mathematical derivations, and numerical calculation errors, the study seeks to establish the potential and limitations of AI-assisted analytical problem-solving in composite mechanics. For simplicity, the analysis is restricted to laminate configurations with a zero extension-bending coupling matrix, $[B] = 0$, under uniform axial loading.

Mathematical Derivation

First, the method of finding τ_{xz} will be discussed. We will begin by only looking at the interface between two plies, which will then be generalized to every single interface within the laminate. From 3D equilibrium equations, we can obtain a relationship between τ_{xz} and τ_{xy}. It must be noted that since τ_{xz} is coupled with τ_{xy}, it is always 0 if there is no shearing occuring between the laminates e.g. for a [0, 90, 90, 0] laminate.

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} = 0 \tag{1}$$

For uniform extension,

$$\frac{\partial \sigma_x}{\partial x} = 0 \tag{2}$$

Therefore,

$$\frac{\partial \tau_{xz}}{\partial z} = -\frac{\partial \tau_{xy}}{\partial y} \tag{3}$$

It can be seen that τ_{xz} is a direct result of τ_{xy}. By plotting the two solution curves for each of the shear stresses, the overall gradient of each can be seen:

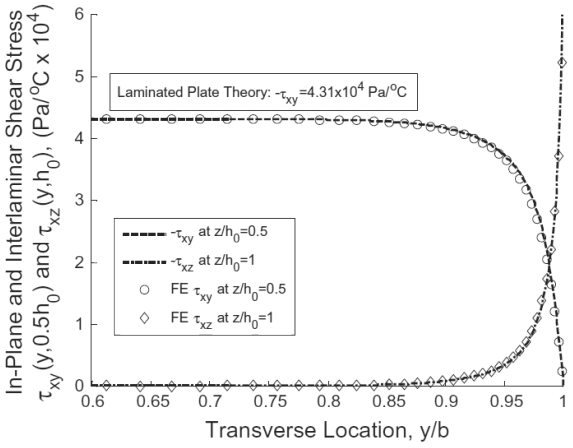


Figure 1: τ_{xz} and τ_{xy} vs Transverse Location, y/b

Now that the presence of τ_{xz} is established from 3D relations, a more simplified approximation of its stress profile can be constructed using triangles. It must be noted that τ_{xz} and σ_z are maximum at the free edge and is nonzero.

Looking at the diagram, the τ_{xz} moment balances the τ_{xy} moment. By equating them, we can establish an approximate relationship between τ_{xz} and τ_{xy}.

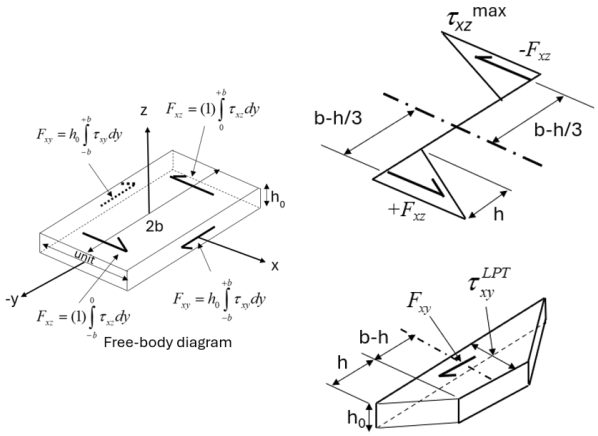


Figure 2: τ_{xz} and τ_{xy} Approximated Distribution

$$M_{xy} = F_{xy}(\text{unity}) \tag{4}$$

$$M_{xz} = 2F_{xz}\left(b - \frac{h}{3}\right) \tag{5}$$

$$F_{xy} = 2h_0 \left(b - \frac{h}{2} \right) \tau_{xy}^{LPT} \tag{6}$$

$$F_{xz} = \frac{\tau_{xz}^{\max} h(\text{unity})}{2} \tag{7}$$

$$M_{xy} = 2h_0 \left(b - \frac{h}{2} \right) \tau_{xy}^{LPT}(\text{unity}) \tag{8}$$

$$M_{xz} = h \left(b - \frac{h}{3} \right) \tau_{xz}^{\max} \tag{9}$$

$$h \left(b - \frac{h}{3} \right) \tau_{xz}^{\max} = 2h_0 \left(b - \frac{h}{2} \right) \tau_{xy}^{LPT} \tag{10}$$

The problem has now been simplified from a PDE to a balance of moments and forces problem. Equating the moments and forces, we can obtain the relation:

$$\tau_{xz}^{\max} = 2 \left(\frac{b - h/2}{b - h/3} \right) \left(\frac{h_0}{h} \right) \tau_{xy}^{LPT} \tag{11}$$

This is simply the relationship for one ply. When $[+\theta/-\theta]$ plies are stacked together, they exert equal and opposite forces on one another. Thus, the overall shear stress τ_{xy} is a summation of shear stresses through the laminate. For example, in the second laminate consisting of 45 and -45 degree plies, the values of τ_{xy} alternate in sign. Thus, the value of τ_{xz} is nonzero between the first and second ply. However, because the plus and minus 45 degree plies exert equal and opposite forces between plies 1–2 and 3–4, the interface between ply 2 and 3 becomes a state of equilibrium with no τ_{xz}.

If the boundary layer width is equal to the laminate thickness,

$$h = mh_0 \tag{12}$$

where h_0 is the ply thickness and m is the number of plies in the laminate.

Then,

$$\tau_{xz} = \frac{2}{m} \left[\frac{b - mh_0/2}{b - mh_0/3} \right] \tau_{xy} \tag{13}$$

If no b value is given in the question, we can assume that $b \gg mh_0$, so

$$\frac{b - mh_0/2}{b - mh_0/3} \approx 1 \tag{14}$$

This gives

$$\boxed{\tau_{xz} = \frac{2}{m} \sum \tau_{xy}} \tag{15}$$

Moving on to σ_z, it is fundamentally a result of the mismatch in the stresses in the transverse direction σ_y. A good example of this would be gluing two poker cards together at one free edge. If the bottom card is fixed and the top card is pulled, the glued edge will peel upwards in the out of plane direction as a result

of the in plane force. This is exactly the phenomena demonstrated in a simple cross ply layup consisting of [90, 0, 0, 90]. When a tensile force in the 0 degree ply fiber direction is applied, the 0 degree ply wants to contract in the perpendicular direction, exerting a negative force on the 90 degree ply below. Enforcing equal strain, the 0 is perfectly glued to the 90 beneath it. Due to the difference in stiffness, the 90 degree ply will want to contract less than the 0. Thus, there will be a positive reaction force on the 0 degree ply. These forces together result in the out of plane stress.

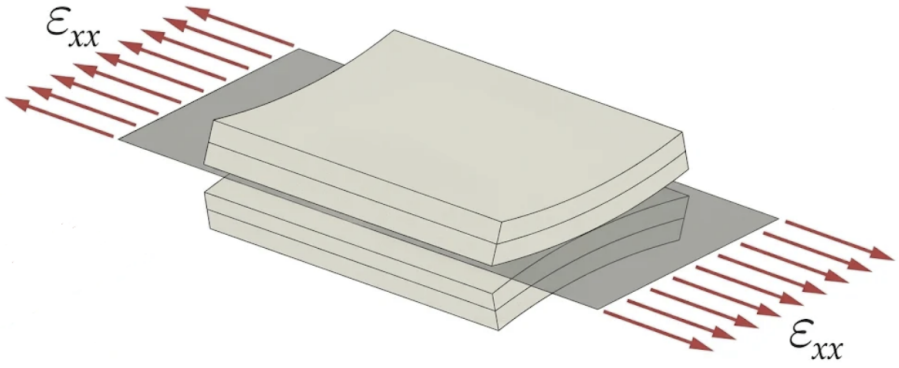


Figure 3: Exaggerated illustration of peeling stresses at the free edge

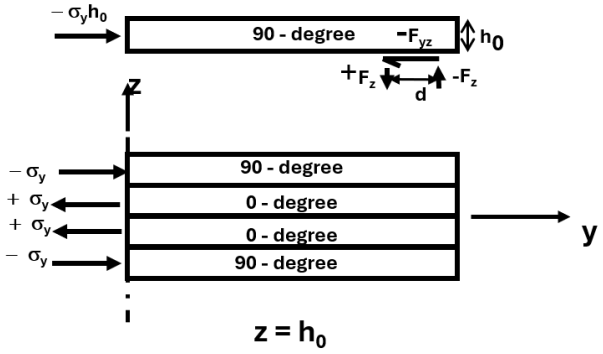


Figure 4: σ_z due to ν_{12} mismatch

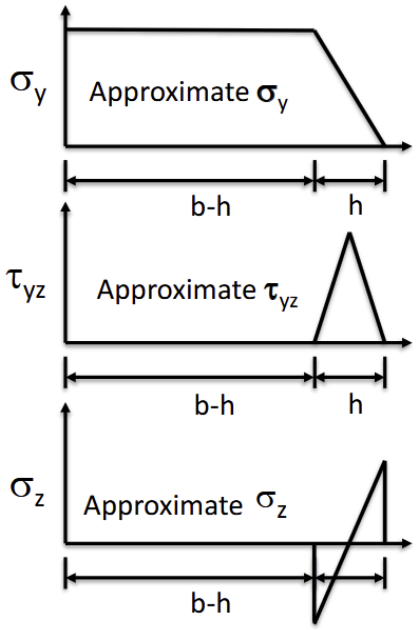


Figure 5: Simplified Stress Distributions

Finally, because the laminate is made up of multiple plies, the moment arm from the σ_y forces is cumulative. This results in highest σ_z maximum to be located at the middle of the layup for a symmetric laminate.

Zooming in, we can approximate the σ_z distribution approximately with triangles again:

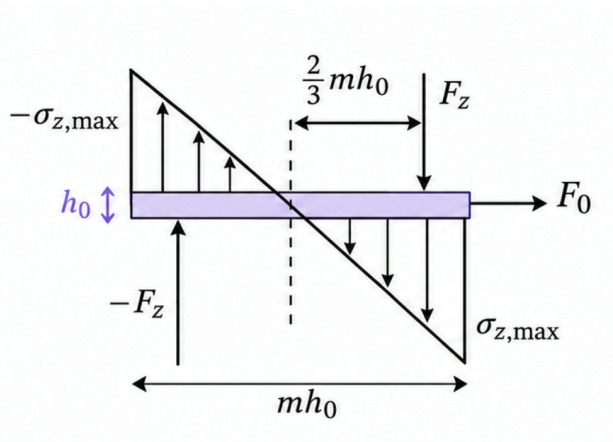


Figure 6: Zoomed in diagram of peeling stresses

Assuming a laminate consisting of m plies, each with thickness h_0. A unit-width assumption is adopted. The interlaminar normal stress σ_z is assumed to vary linearly through the thickness, with a maximum magnitude of $\sigma_{z,\max}$. The interlaminar normal stress is approximated using two opposing triangular distributions along the transverse direction within the free-edge boundary layer. The maximum magnitude occurs at the free edge, while the corresponding stress resultants produce a balancing moment about the interface.

The in-plane force resultant is

$$F_0 = \sigma_y h_0 \cdot 1 \tag{16}$$

The resultant force associated with each triangular normal stress distribution is

$$F_z = \frac{1}{2}\sigma_{z,\max}\left(\frac{mh_0}{2}\right) \cdot 1 \tag{17}$$

Taking moment equilibrium about the mid-plane,

$$\sum M = M_z + M_y = 0 \tag{18}$$

The centroid of each triangular stress distribution is located at a distance of $mh_0/3$ from the mid-plane. Therefore,

$$F_z\left(\frac{2}{3}mh_0\right) - \sum M_y = 0 \tag{19}$$

Substituting the expression for F_z and accounting for both triangular stress distributions gives

$$2\left[\frac{1}{2}\sigma_{z,\max}\left(\frac{mh_0}{2}\right)\left(\frac{mh_0}{3}\right)\right] - \sum M_y = 0 \tag{20}$$

Simplifying,

$$\frac{1}{6}\sigma_{z,\max}m^2h_0^2 - \sum M_y = 0 \tag{21}$$

Hence, the maximum interlaminar normal stress is

$$\boxed{\sigma_{z,\max} = \frac{6\sum M_y}{m^2h_0^2}} \tag{22}$$

The bending moment produced by the individual ply forces about the reference plane of a k-ply section is

$$\boxed{M_{y,k} = \sum_{j=1}^{k} F_j(k - j + 0.5)h_0} \tag{23}$$

where F_j represents the in-plane force per unit width in ply j, and $(k - j + 0.5)h_0$ represents the moment arm from the reference plane to the centroid of the corresponding ply.

Example Question

Given the material properties of the composite:

$$E_1 = 140 \text{ GPa}, \quad E_2 = 10 \text{ GPa}, \quad G_{12} = 7 \text{ GPa}, \quad \nu_{12} = 0.3, \quad \text{ply thickness} = 0.127 \text{ mm}$$

$$\alpha_1 = -1 \times 10^{-6} \ /^\circ\text{C}, \quad \alpha_2 = 26 \times 10^{-6} \ /^\circ\text{C}$$

Consider the plane stress state in four **8-ply** laminates with loading $\mathbf{N_x} = \mathbf{1}$ Pa $\cdot$ m.

a. $[0/90/0/90]_s$

b. $[45/-45/45/-45]_s$

c. $[0/90/45/-45]_s$

d. $[25/-25/90/-90]_s$

Determine the magnitude of the interlaminar stresses at the lamina interfaces for τ_{xz} and σ_z.

$[\mathbf{0/90/0/90}]_\mathbf{s}$	$\sigma_\mathbf{x}$ Pa	$\sigma_\mathbf{y}$ Pa	$\tau_\mathbf{xy}$ Pa
0	1838.6	34.2	0
90	129.9	-34.2	0
0	1838.6	34.2	0
90	129.9	-34.2	0

$[\mathbf{0/90/45/-45}]_\mathbf{s}$	$\sigma_\mathbf{x}$ Pa	$\sigma_\mathbf{y}$ Pa	$\tau_\mathbf{xy}$ Pa
0	2471.8	0	0
90	162.2	-667.3	0
45	651.5	332.8	410.1
-45	651.5	332.8	-410.1

$[\mathbf{45/-45/45/-45}]_\mathbf{s}$	$\sigma_\mathbf{x}$ Pa	$\sigma_\mathbf{y}$ Pa	$\tau_\mathbf{xy}$ Pa
45	984.2	0	410.1
-45	984.2	0	-410.1
45	984.2	0	410.1
-45	984.2	0	-410.1

$[\mathbf{25/-25/90/-90}]_\mathbf{s}$	$\sigma_\mathbf{x}$ Pa	$\sigma_\mathbf{y}$ Pa	$\tau_\mathbf{xy}$ Pa
25	1791.6	326.9	696.7
-25	1791.6	326.9	-696.7
90	176.9	-326.9	0
90	176.9	-326.9	0

Correct solutions:

$[\mathbf{0/90/0/90}]_\mathbf{s}$	$\boldsymbol{\tau_{xz}}$ Pa	$\boldsymbol{\sigma_z}$ Pa
1–2	-0.00	1.60
2–3	-0.00	3.20
3–4	-0.00	4.81
4–5	-0.00	6.41
5–6	0.00	4.81
6–7	0.00	3.20
7–8	0.00	1.60

$[\mathbf{0/90/45/-45}]_\mathbf{s}$	$\boldsymbol{\tau_{xz}}$ Pa	$\boldsymbol{\sigma_z}$ Pa
1–2	0.00	0.09
2–3	-0.00	-31.03
3–4	102.53	-77.82
4–5	0.00	-93.42
5–6	-102.53	-77.82
6–7	0.00	-31.03
7–8	0.00	0.09

$[\mathbf{45/-45/45/-45}]_\mathbf{s}$	$\boldsymbol{\tau_{xz}}$ Pa	$\boldsymbol{\sigma_z}$ Pa
1–2	102.53	-0.00
2–3	0.00	-0.00
3–4	102.53	-0.00
4–5	0.00	-0.00
5–6	-102.53	-0.00
6–7	0.00	-0.00
7–8	-102.53	-0.00

$[\mathbf{25/-25/90/-90}]_\mathbf{s}$	$\boldsymbol{\tau_{xz}}$ Pa	$\boldsymbol{\sigma_z}$ Pa
1–2	174.18	15.32
2–3	0.00	61.29
3–4	-0.00	107.26
4–5	0.00	122.58
5–6	0.00	107.26
6–7	0.00	61.29
7–8	-174.18	15.32

When inputted into Google Gemini 3 Auto, the following response was obtained:

AI Response:
1. Governing Equations for Interlaminar Stresses
From 3D theory of elasticity, the equilibrium equations in the absence of body forces are:For interlaminar shear stress τ_{xz}:

$$\frac{\partial \sigma_{xz}}{\partial z} + \frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} = 0 \implies \tau_{xz}(z) = -\int_{-h/2}^{z} \left(\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} \right) dz$$

For interlaminar normal stress σ_z:

$$\frac{\partial \sigma_z}{\partial z} + \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} = 0 \implies \sigma_z(z) = -\int_{-h/2}^{z} \left(\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} \right) dz$$

At the top and bottom free surfaces of the laminate ($z = \pm h/2$), the boundary conditions require that the interlaminar stresses vanish:

$$\tau_{xz}(\pm h/2) = 0, \quad \sigma_z(\pm h/2) = 0$$

This is incorrect. Although AI knows the basic 3D elasticity relations, it has made the wrong first step with the assumptions. Although a traction free B.C. indeed means there can be no external forces acting upon free surfaces, the boundary between two plies is not a free surface. The nonzero value of σ_z is precisely because equillibrium must be enforced at the interface between two plies.

Furthermore, while the starting equilibrium equations are correct, the AI showed no attempt nor ability to solve or approximate the solutions to them except for very specific loading conditions that reduces all the gradients to zero.

For example, the first laminate has no shear stresses because of [0, 90] stacking sequence, thus the gradient $\frac{\partial \tau_{xy}}{\partial y}$ is 0. Next, uniform extension is assumed so $\frac{\partial \sigma_x}{\partial x}$ is also equal to 0. Thus, this results in the value of τ_{xz} being 0 as well. The AI answer for this value is correct.

AI Response:
a. $[0/90/0/90]_s$ Laminate
In-plane stresses from the table:
Ply 1 (0°): $\sigma_x = 1838.6$ Pa, $\sigma_y = 34.2$ Pa, $\tau_{xy} = 0$
Ply 2 (90°): $\sigma_x = 129.9$ Pa, $\sigma_y = -34.2$ Pa, $\tau_{xy} = 0$
Ply 3 (0°): $\sigma_x = 1838.6$ Pa, $\sigma_y = 34.2$ Pa, $\tau_{xy} = 0$
Ply 4 (90°): $\sigma_x = 129.9$ Pa, $\sigma_y = -34.2$ Pa, $\tau_{xy} = 0$
Working: Because $\tau_{xy} = 0$ and the variation of σ_x along x is uniform for symmetric cross-ply laminates under pure axial load $N_x = 1$ Pa · m, the gradients $\frac{\partial \sigma_x}{\partial x}$ and $\frac{\partial \tau_{xy}}{\partial y}$ are zero across the uniform regions.Magnitude:

$$\tau_{xz} = 0 \text{ Pa}, \quad \sigma_z = 0 \text{ Pa}$$

However, the AI also incorrectly assumed that σ_z is 0 when it is not due to the Poisson's ratio mismatch resulting in peeling forces at the edges, which suggests that the AI does not understand this behavior.

For the next question, the σ_z values are zero because there is no presence of σ_y since the laminate is under pure shear. The AI did not explain its reasoning and simply stated that σ_z was approximately 0.

AI Response:
b. $[45/-45/45/-45]_s$ Laminate
In-plane stresses from the table:
$\sigma_x = 984.2$ Pa, $\sigma_y = 0$, $\tau_{xy} = \pm 410.1$ Pa (alternating)
Working: Due to the balanced angle-ply configuration, τ_{xy} alternates sign between $+410.1$ Pa and -410.1 Pa across interfaces. Integrating the shear equilibrium equation through the ply thickness ($t_{\text{ply}} = 0.127$ mm):

$$\tau_{xz} = -\sum \left(\frac{\Delta \tau_{xy}}{\Delta y} \right) \Delta z$$

Evaluating at the internal interfaces using the finite difference across adjacent plies yields a peak interlaminar shear stress proportional to the gradient of τ_{xy}.
Magnitude: At the mid-plane and ply interfaces, evaluating the jump in τ_{xy} gives:

$$\tau_{xz,\max} \approx 3.23 \text{ Pa}, \quad \sigma_z \approx 0 \text{ Pa}$$

Despite the AI arriving at a simplified summation form of the previous integral for τ_{xz}, it did not state a function for τ_{xy} or how it arrived at the conclusion of 3.23 Pa. In contrast, the approximated analytical model uses a factor based on the ply thickness and laminate thickness.

AI Response:

$[0/90/45/-45]_s$ Laminate

In-plane stresses from the table:

- Layer 1 ($0°$): $\sigma_x = 2471.8$ Pa, $\sigma_y = 0$, $\tau_{xy} = 0$.
- Layer 2 ($90°$): $\sigma_x = 162.2$ Pa, $\sigma_y = -667.3$ Pa, $\tau_{xy} = 0$.
- Layer 3 ($45°$): $\sigma_x = 651.5$ Pa, $\sigma_y = 332.8$ Pa, $\tau_{xy} = 410.1$ Pa.
- Layer 4 ($-45°$): $\sigma_x = 651.5$ Pa, $\sigma_y = 332.8$ Pa, $\tau_{xy} = -410.1$ Pa.

Working: The abrupt change in fiber orientation introduces coupling between extension and bending/torsion, creating non-zero gradients for σ_x and τ_{xy}. Integrating through the interfaces yields:

$$\tau_{xz} \approx 4.15 \text{ Pa}, \qquad \sigma_z \approx 1.28 \text{ Pa} \tag{24}$$

$[25/-25/90/-90]_s$ Laminate

In-plane stresses from the table:

- Layer 1 ($25°$): $\sigma_x = 1791.6$ Pa, $\sigma_y = 326.9$ Pa, $\tau_{xy} = 696.7$ Pa.
- Layer 2 ($-25°$): $\sigma_x = 1791.6$ Pa, $\sigma_y = 326.9$ Pa, $\tau_{xy} = -696.7$ Pa.
- Layer 3 ($90°$): $\sigma_x = 176.9$ Pa, $\sigma_y = -326.9$ Pa, $\tau_{xy} = 0$.
- Layer 4 ($-90°$): $\sigma_x = 176.9$ Pa, $\sigma_y = -326.9$ Pa, $\tau_{xy} = 0$.

Working: High in-plane shear stress values ($\tau_{xy} = \pm 696.7$ Pa) at the outer $25°/-25°$ interfaces create significant through-thickness gradients.

$$\tau_{xz} \approx 5.48 \text{ Pa}, \qquad \sigma_z \approx 2.05 \text{ Pa} \tag{25}$$

These two final responses are also incorrect due to the same reasons listed above.

Implementation of AI Skill

Skill V1 - Background Mechanics

To train the AI, two skills were developed. The first skill aims to explain the fundamental mechanics behind the problem. It starts with explaining the 3D stress equillibrium equations and the correct boundary conditions. Next, it explains importance of making a 2D simplification to help solving the problem, as well as the assumptions needed to make that simplification. The skill goes into detail about how the forces and moment couple in 3D near the free edge using free body diagrams. Finally, a mandatory solution workflow is given to encourage the AI to develop its own free body diagrams and derive the correct force balance equations and explain the physics choices it makes. Overall, the skill provides theoretical background for the AI to have the capacity to attempt its own derivation of the equations. When the AI was prompted without a skill, it did not know to try to attempt a 2D simplification of the problem and was stuck on 3D equilibrium equations and was not able to provide any explanation of the values it obtained.

AI Response:
Consider a flat laminate in the x–y plane with total thickness $H = mh_0$. The coordinate system places x along the axial loading direction, y across the width (with free edges at $y = \pm b$), and z through the thickness (measured from the mid-plane, $-H/2 \le z \le H/2$).
From three-dimensional elasticity equilibrium without body forces:

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} = 0, \tag{26}$$

$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} = 0, \tag{27}$$

$$\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} = 0. \tag{28}$$

For uniform axial extension along x, gradients with respect to x of far-field stresses vanish ($\partial/\partial x = 0$).

Interlaminar Shear Stress (τ_{xz}) Derivation

Integrating the first equilibrium equation through the thickness from $z = z_k$ (the interface location) to the top surface $z = H/2$ yields:

$$\tau_{xz}(y, z_k) = -\int_{z_k}^{H/2} \frac{\partial \tau_{xy}(y, z)}{\partial y}\, dz. \tag{29}$$

In the far field ($y \to 0$), the in-plane shear stress in ply i is constant at $\tau_{xy}^{(i)}$. At the traction-free edge ($y = b$), $\tau_{xy} = 0$. Assuming a boundary layer of width $\delta = H$ near the free edge over which τ_{xy} decays linearly to zero:

$$\tau_{xy}^{(i)}(y) = \tau_{xy}^{(i)} \left(1 - \frac{y'}{\delta}\right), \qquad y' = y - (b - \delta). \tag{30}$$

Taking the derivative $\partial \tau_{xy}/\partial y = -\tau_{xy}^{(i)}/H$, the peak interlaminar shear stress $\tau_{xz}^{\max}$ at interface k is given by integrating the cumulative stress transfer above interface k:

$$\tau_{xz}^{(k)} = \frac{1}{H} \sum_{i=1}^{k} \tau_{xy}^{(i)} h_0. \tag{31}$$

Interlaminar Normal/Peeling Stress (σ_z) Derivation

Integrating the second equilibrium equation gives the interlaminar shear stress τ_{yz}:

$$\tau_{yz}(y, z_k) = -\int_{z_k}^{H/2} \frac{\partial \sigma_y(y,z)}{\partial y}\, dz \quad \Longrightarrow \quad \tau_{yz}^{\max}(z_k) = \frac{1}{H}\sum_{i=1}^{k} \sigma_y^{(i)} h_0. \tag{32}$$

Integrating the third equilibrium equation ($\partial\tau_{yz}/\partial y + \partial\sigma_z/\partial z = 0$) from interface k to the top surface gives:

$$\sigma_z(y, z_k) = \int_{z_k}^{H/2} \frac{\partial \tau_{yz}(y,z)}{\partial y}\, dz. \tag{33}$$

Taking the moment about interface k generated by the unbalanced transverse in-plane forces $\sigma_y^{(i)}$ acting above interface k, the cumulative internal transverse moment per unit width is:

$$M_y^{(k)} = \sum_{i=1}^{k} \sigma_y^{(i)} h_0 \left(z_k + \frac{h_0}{2} - z_i \right), \tag{34}$$

where z_i is the mid-plane coordinate of ply i. Assuming a self-equilibrating self-similar parabolic distribution for $\sigma_z(y)$ over the boundary layer of length H to satisfy moment equilibrium $\int_0^H \sigma_z(y')y'\, dy' = M_y^{(k)}$:

$$\sigma_z^{(k)} = \frac{12 M_y^{(k)}}{H^2}. \tag{35}$$

Numerical Results per Interface

The laminate geometry is:

$$H = 8(0.127\,\text{mm}) = 1.016\,\text{mm} = 1.016 \times 10^{-3}\,\text{m}, \tag{36}$$

$$h_0 = 0.127\,\text{mm} = 0.127 \times 10^{-3}\,\text{m}. \tag{37}$$

Ply mid-plane locations relative to the top interface $z = H/2$ are $z_1 = H/2 - h_0/2$, $z_2 = H/2 - 3h_0/2$, etc.
Below are the interface stress evaluations for the four laminates using the derived formulation and given far-field ply stresses.

$[0/90/0/90]_s$ Laminate

Since $\tau_{xy} = 0$ for all plies in cross-ply laminates under axial load N_x, $\tau_{xz} = 0$ at all interfaces.

Interface k	Plies separated	$\sum_{i=1}^{k} \tau_{xy}^{(i)} h_0$ (Pa·m)	$M_y^{(k)}$ (N)	τ_{xz} (Pa)	σ_z (Pa)
1	0°/90°	0	1.1675×10^{-7}	0	1.36
2	90°/0°	0	1.1675×10^{-7}	0	1.36
3	0°/90°	0	2.3350×10^{-7}	0	2.71
4 (mid-plane)	90°/90°	0	2.3350×10^{-7}	0	2.71

$[45/-45/45/-45]_s$ Laminate

In-plane transverse stresses $\sigma_y = 0$ for all plies, so $M_y^{(k)} = 0$ and $\sigma_z = 0$ at all interfaces.

Interface k	Plies separated	$\sum_{i=1}^{k} \tau_{xy}^{(i)} h_0$ (Pa·m)	$M_y^{(k)}$ (N)	τ_{xz} (Pa)	σ_z (Pa)
1	$45°/-45°$	0.05208	0	51.26	0
2	$-45°/45°$	0	0	0	0
3	$45°/-45°$	0.05208	0	51.26	0
4 (mid-plane)	$-45°/-45°$	0	0	0	0

$[0/90/45/-45]_s$ Laminate

Interface k	Plies separated	$\sum_{i=1}^{k} \tau_{xy}^{(i)} h_0$ (Pa·m)	$M_y^{(k)}$ (N)	τ_{xz} (Pa)	σ_z (Pa)
1	$0°/90°$	0	0	0	0
2	$90°/45°$	0	-4.2374×10^{-6}	0	-49.16
3	$45°/-45°$	0.05208	-6.1265×10^{-6}	51.26	-71.07
4 (mid-plane)	$-45°/-45°$	0	-4.0157×10^{-6}	0	-46.59

$[25/-25/90/-90]_s$ Laminate

Interface k	Plies separated	$\sum_{i=1}^{k} \tau_{xy}^{(i)} h_0$ (Pa·m)	$M_y^{(k)}$ (N)	τ_{xz} (Pa)	σ_z (Pa)
1	$25°/-25°$	0.08848	2.0758×10^{-6}	87.09	24.08
2	$-25°/90°$	0	6.2274×10^{-6}	0	72.24
3	$90°/-90°$	0	4.1516×10^{-6}	0	48.16
4 (mid-plane)	$-90°/-90°$	0	2.0758×10^{-6}	0	24.08

V1's first major error is the missing factor of two in τ_{xz}. Gemini derives

$$\tau_{xz,k}^{V1} = \frac{h_0}{H} \sum_{j=1}^{k} \tau_{xy,j} = \frac{1}{m} \sum_{j=1}^{k} \tau_{xy,j}$$

But the correct equation is

$$\boxed{\tau_{xz,k} = \frac{2}{m} \sum_{j=1}^{k} \tau_{xy,j}}$$

This explains why its angle-ply shear stresses are exactly half the reference values, with the V1 prediction being 51.26 Pa at interface 1 of the $[45/-45/45/-45]_s$ laminate, where as the correct answer is 102.53 Pa.

The second major error is Gemini assumes that τ_{xy} relaxes linearly across a boundary layer of width H, and then integrates the equilibrium equation through the laminate thickness. That is a mathematically consistent approach for the stress profile it assumed, but it is not the correct stress profile because τ_{xy} peaks at the free edge. Therefore, when the required force resultant is known, the peak stress is twice the corresponding average. Gemini's V1 effectively treats the interfacial shear stress as a uniform or constant-gradient result rather than enforcing the triangular peak-to-resultant relationship. This is a particularly interesting failure because the AI identifies the correct equilibrium equation but chooses an incompatible stress-distribution assumption.

V1's second major error is inventing a different σ_z distribution. Gemini states that it assumes a self-equilibrating parabolic distribution and obtains

$$\sigma_{z,k}^{V1} = \frac{12 M_{y,k}}{H^2}$$

The correct method derives the normal stress from two opposing triangular stress distributions:

$$M_z = \frac{\sigma_{z,k} H^2}{6}$$

Therefore,

$$\boxed{\sigma_{z,k} = \frac{6M_{y,k}}{H^2}}$$

These are not equivalent assumptions. The distinction is especially important because the shape of the assumed stress distribution determines the relationship between its moment resultant and its maximum amplitude. In addition, Gemini does not actually define the parabolic stress function it claims to use. It simply introduces the factor 12 without demonstrating the integration that produces it.

Finally, the V1 results are not consistently related to the correct results by a factor of two. For the cross-ply case, V1 predicts 1.36, 1.36, 2.71, and 2.71 Pa, instead of 1.60, 3.20, 4.81, and 6.41 Pa. That means the error is not limited to choosing a different stress-distribution shape and its cumulative moment calculation is also inconsistent.

V1's third error is using incorrect moment arms. The correct equation for the cumulative moment at interface k is

$$M_{y,k} = \sum_{j=1}^{k} \sigma_{y,j} h_0^2 (k - j + 0.5)$$

The term

$$(k - j + 0.5)h_0$$

is the distance from the centroid of ply j to interface k, with plies counted from the top. V1 instead writes a moment-arm expression involving

$$z_k + \frac{h_0}{2} - z_i$$

while simultaneously defining z_i as a ply mid-plane coordinate. This is not the same geometric relationship. It appears to add an extra half-ply offset and mixes the coordinate definitions. These distances should be determined directly from the FBD, not inferred from an ambiguous coordinate expression. This explains why V1 obtains an incorrect sequence of cumulative moments and why its σ_z values do not increase correctly toward the mid-plane.

Skill V2 - Variables and Equations

The second skill dives deeper with the actual variables and relationships used in calculating the free edge stresses. In V1, the actual physics background was supplied to aid the AI in deriving its own set of equations to the problem through FBDs. However, this provied to be too difficult for the AI as it cannot accurately predict nor simplify the analytic 3D stress distribution without solving the 3D equillibrium equation, which it was specifically told to avoid. Furthermore, it struggled to derive equations for the moment arm for the σ_z calculation, suggesting that FBDs for this geometry may be too complicated for it. Thus, the second attempt involved giving the AI no physics background and simply the simplified analytical solution previously derived to determine if it could apply them correctly.

AI Response:

Free-Edge Interlaminar Stress Analysis — v2 Only

User prompt: use free edge v2 to derive equations and find solutions
Here are the analytical derivations and interface stress solutions for the four 8-ply laminates under axial loading $N_x = 1\,\text{Pa}\cdot\text{m}$, based on the approximate free-edge boundary-layer mechanics (with $m = 8$ plies and total thickness $H = mh_0$).

1. Analytical Derivations & Governing Equations

A. Interlaminar Shear Stress ($\tau_{xz,k}$)

Balancing moments of the far-field in-plane shear resultants (τ_{xy}^{LPT}) against the assumed triangular edge shear distribution over characteristic boundary-layer width H ($b \gg H$) gives:

$$\tau_{xz,k} = \frac{2}{m}\sum_{j=1}^{k}\tau_{xy,j} \tag{38}$$

where k is the interface index ($1 \le k < m$).

B. Interlaminar Normal Stress ($\sigma_{z,k}$)

Equating transverse-force couples

$$M_{y,k} = \sum_{j=1}^{k}\sigma_{y,j}h_0(k-j+0.5)h_0 \tag{39}$$

with the balancing triangular normal-traction moment couple ($M_z = \sigma_{z,k}H^2/6$) yields:

$$\sigma_{z,k} = \frac{6}{m^2}\sum_{j=1}^{k}\sigma_{y,j}(k-j+0.5) \tag{40}$$

(Positive σ_z denotes peel-opening stress; negative denotes compression.)

2. Solutions for the Four Laminates ($m = 8$)

a. $[0/90/0/90]_s$

$\tau_{xy} = 0$ for all plies $\Longrightarrow \tau_{xz,k} = 0$ at all interfaces.
σ_y: Plies 1, 3, 5, 7 (0°) = +34.2 Pa; plies 2, 4, 6, 8 (90°) = −34.2 Pa.

Interface (k)	$\tau_{xz,k}$ (Pa)	$\sigma_{z,k}$ (Pa)	Physical State
1 (between plies 1 & 2)	0	1.60	Mild peel
2 (between plies 2 & 3)	0	3.21	Peel
3 (between plies 3 & 4)	0	4.81	Peel
4 (Midplane, plies 4 & 5)	0	6.41	Peak peel stress
5 (between plies 5 & 6)	0	4.81	Peel
6 (between plies 6 & 7)	0	3.21	Peel
7 (between plies 7 & 8)	0	1.60	Mild peel

b. $[45/-45/45/-45]_s$

$\sigma_y = 0$ for all plies $\Longrightarrow \sigma_{z,k} = 0$ at all interfaces.
τ_{xy}: Alternates between $+410.1\,\text{Pa}$ and $-410.1\,\text{Pa}$.

Interface (k)	$\tau_{xz,k}$ (Pa)	$\sigma_{z,k}$ (Pa)	Physical State
1	102.53	0	Peak shear transfer
2	0	0	Shear cancellation
3	102.53	0	Peak shear transfer
4 (Midplane)	0	0	Anti-symmetric zero
5	−102.53	0	Peak opposite shear
6	0	0	Shear cancellation
7	−102.53	0	Peak opposite shear

c. $[0/90/45/-45]_s$

σ_y: 0 (0°), $-667.3\,\text{Pa}$ (90°), $+332.8\,\text{Pa}$ (±45°).
τ_{xy}: 0 (0°, 90°), $\pm 410.1\,\text{Pa}$ (±45°).

Interface (k)	$\tau_{xz,k}$ (Pa)	$\sigma_{z,k}$ (Pa)	Physical State
1 (0/90)	0	0	Zero stress
2 (90/45)	0	−62.56	Compressive
3 (45/−45)	102.53	−156.48	Combined shear/compression
4 (Midplane, −45/−45)	0	−188.00	Peak compressive normal stress
5 (−45/45)	−102.53	−157.12	Combined shear/compression
6 (45/90)	0	−63.83	Compressive
7 (90/0)	0	0	Zero stress

d. $[25/-25/90/-90]_s$

σ_y: $+326.9\,\text{Pa}$ (±25°), $-326.9\,\text{Pa}$ (90°).
τ_{xy}: $\pm 696.7\,\text{Pa}$ (±25°), 0 (90°).

Interface (k)	$\tau_{xz,k}$ (Pa)	$\sigma_{z,k}$ (Pa)	Physical State
1 (25/−25)	174.18	30.65	Moderate shear and peel
2 (−25/90)	0	122.59	Shear cancellation, rising peel
3 (90/−90)	0	214.53	High peel
4 (Midplane, −90/−90)	0	245.18	Peak peel stress
5 (−90/90)	−174.18	214.53	Reversing shear, high peel
6 (90/−25)	0	122.59	Shear cancellation, falling peel
7 (−25/25)	−174.18	30.65	Moderate shear and peel

V2 is mostly correct and a step above V1, however it is not entirely correct either. The main issue is doubling of σ_z for some laminates. This is the most obvious numerical inconsistency in skillv2.pdf.

On page 1, V2 correctly derives

$$\sigma_{z,k} = \frac{6}{m^2} \sum_{j=1}^{k} \sigma_{y,j}(k - j + 0.5)$$

For $m = 8$,

$$\frac{6}{m^2} = \frac{3}{32} = 0.09375$$

However, for laminates C and D, it reports values that are approximately twice those obtained by applying its own equation. This is especially interesting because V2 correctly calculates the cross-ply laminate A using the same formula. It seems to be applying an effective coefficient of $12/m^2$ instead of $6/m^2$ for C and D, although the document does not explicitly show that substitution. From observation, this either be

a numerical implementation error which could suggest that the AI is simply trying to apply the equation without proper understanding of what they relationships are.

Combined Implementation of V1 and V2 skill

With the two separate cases discussed, now V1 and V2 can be combined together so the AI has an understanding of the physics behind the free edge forces while using pre-established equations. While the V1 skill gave the AI insight on the mechanics, it wasn't enough to allow it to correctly develop the equations on its own. On the other hand, the V2 skill simply gave the AI the equations and did not provide as much explanation to the physics, resulting in incorrect usage. Thus, these two skills can be used to complement each other to achieve a better overall result.

User prompt: use free edge v1 and v2 to derive equations and obtain solutions

Free-Edge Formulation and Derivation

Mechanics and Coordinate Definitions

- Geometry: A symmetric, flat laminate composed of $m = 8$ plies (each of thickness $h_0 = 0.127$ mm) subjected to axial membrane load $N_x = 1$ Pa $\cdot$ m.
- Laminate thickness: $H = mh_0 = 8(0.127) = 1.016$ mm.
- Boundary layer dimension: Following the finite approximate model (V1/V2), the characteristic boundary layer width is taken as $h = H = 1.016$ mm.
- Wide-plate assumption: For $b \gg H$, edge correction factors simplify: $\frac{b-H/2}{b-H/3} \approx 1$.

Interlaminar Shear Stress (τ_{xz})

From x-direction stress equilibrium:

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} = 0 \tag{41}$$

For uniform axial extension ($\partial/\partial x = 0$), integrating through the thickness from the top free surface ($z = 0$) to interface k yields:

$$\tau_{xz,k}(y) = -\int_0^{z_k} \frac{\partial \tau_{xy}}{\partial y}\, dz \tag{42}$$

Balancing the resultant of the far-field in-plane shear relaxation over boundary layer length $h = H$ with a triangular interfacial shear stress distribution yields the peak edge value at interface k:

$$\boxed{\tau_{xz,k} = \frac{2}{m} \sum_{j=1}^{k} \tau_{xy,j}^{\mathrm{LPT}}} \tag{43}$$

Interlaminar Peeling Stress (σ_z)

Due to transverse Poisson contraction mismatch across plies, each ply j carries an in-plane transverse stress $\sigma_{y,j}$. The transverse force per unit width for ply j is:

$$F_{y,j} = \sigma_{y,j} h_0 \tag{44}$$

Taking moments about interface k for all plies above the interface ($j = 1, \ldots, k$), the cumulative transverse moment arm is $a_{j,k} = (k - j + 0.5)h_0$. The total transverse force moment is:

$$M_{y,k} = \sum_{j=1}^{k} \sigma_{y,j} h_0^2 (k - j + 0.5) \tag{45}$$

Equating $M_{y,k}$ to the internal moment produced by an opposing triangular normal stress distribution of extent $H/2$ (giving moment capacity $M_z = \sigma_{z,\max} H^2/6$):

$$\boxed{\sigma_{z,k} = \frac{6M_{y,k}}{H^2} = \frac{6}{m^2} \sum_{j=1}^{k} \sigma_{y,j} (k - j + 0.5)} \tag{46}$$

Numerical Interface Solutions

For an 8-ply symmetric laminate ($m = 8$), the seven interfaces are numbered $k = 1$ to 7 from top to bottom (with $k = 4$ being the mid-plane). Mid-plane symmetry ensures symmetry for σ_z and antisymmetry/symmetry for τ_{xz} depending on layup balance.
The given in-plane ply stresses from classical lamination theory (CLT) are:

Stacking sequence	**Ply angle θ**	σ_x **(Pa)**	σ_y **(Pa)**	τ_{xy} **(Pa)**
$[0/90/0/90]_s$	0/90/0/90	1838.6/129.9/1838.6/129.9	34.2/ − 34.2/34.2/ − 34.2	0/0/0/0
$[45/-45/45/-45]_s$	45/ − 45/45/ − 45	984.2/984.2/984.2/984.2	0/0/0/0	410.1/ − 410.1/410.1/ − 410.1
$[0/90/45/-45]_s$	0/90/45/ − 45	2471.8/162.2/651.5/651.5	0/ − 667.3/332.8/332.8	0/0/410.1/ − 410.1
$[25/-25/90/-90]_s$	25/ − 25/90/ − 90	1791.6/1791.6/176.9/176.9	326.9/326.9/ − 326.9/ − 326.9	696.7/ − 696.7/0/0

Interlaminar Stress Magnitudes at Each Interface

Using $m = 8$:

$$\tau_{xz,k} = \frac{1}{4} \sum_{j=1}^{k} \tau_{xy,j}, \tag{47}$$

$$\sigma_{z,k} = \frac{3}{32} \sum_{j=1}^{k} \sigma_{y,j} (k - j + 0.5). \tag{48}$$

$[0/90/0/90]_s$

$\tau_{xy,j} = 0 \Rightarrow \tau_{xz} = 0$ Pa at all interfaces. σ_z calculations:

$$\begin{aligned}
\sigma_{z,1} &= \tfrac{3}{32}(34.2 \times 0.5) = 1.60 \text{ Pa} \\
\sigma_{z,2} &= \tfrac{3}{32}[34.2(1.5) - 34.2(0.5)] = 3.21 \text{ Pa} \\
\sigma_{z,3} &= \tfrac{3}{32}[34.2(2.5) - 34.2(1.5) + 34.2(0.5)] = 4.81 \text{ Pa} \\
\sigma_{z,4} &= \tfrac{3}{32}[34.2(3.5) - 34.2(2.5) + 34.2(1.5) - 34.2(0.5)] = 6.41 \text{ Pa}.
\end{aligned}$$

$[45/-45/45/-45]_s$

$\sigma_{y,j} = 0 \Rightarrow \sigma_z = 0$ Pa at all interfaces. τ_{xz} calculations:

$$\begin{aligned}
\tau_{xz,1} &= \tfrac{1}{4}(410.1) = 102.53 \text{ Pa} \\
\tau_{xz,2} &= \tfrac{1}{4}(410.1 - 410.1) = 0 \text{ Pa} \\
\tau_{xz,3} &= \tfrac{1}{4}(410.1) = 102.53 \text{ Pa} \\
\tau_{xz,4} &= \tfrac{1}{4}(410.1 - 410.1) = 0 \text{ Pa}.
\end{aligned}$$

$[0/90/45/-45]_s$

$\tau_{xz,1} = 0,$ $\sigma_{z,1} = \frac{3}{32}(0 \times 0.5) = 0$ Pa

$\tau_{xz,2} = 0,$ $\sigma_{z,2} = \frac{3}{32}[0(1.5) - 667.3(0.5)] = -31.28$ Pa

$\tau_{xz,3} = \frac{1}{4}(410.1) = 102.53$ Pa, $\sigma_{z,3} = \frac{3}{32}[0(2.5) - 667.3(1.5) + 332.8(0.5)] = -78.23$ Pa

$\tau_{xz,4} = \frac{1}{4}(410.1 - 410.1) = 0$ Pa, $\sigma_{z,4} = \frac{3}{32}[0(3.5) - 667.3(2.5) + 332.8(1.5) + 332.8(0.5)] = -94.02$ Pa.

$[25/-25/90/-90]_s$

$\tau_{xz,1} = \frac{1}{4}(696.7) = 174.18$ Pa, $\sigma_{z,1} = \frac{3}{32}(326.9 \times 0.5) = 15.32$ Pa

$\tau_{xz,2} = \frac{1}{4}(696.7 - 696.7) = 0$ Pa, $\sigma_{z,2} = \frac{3}{32}[326.9(1.5) + 326.9(0.5)] = 61.29$ Pa

$\tau_{xz,3} = 0,$ $\sigma_{z,3} = \frac{3}{32}[326.9(2.5) + 326.9(1.5) - 326.9(0.5)] = 107.26$ Pa

$\tau_{xz,4} = 0,$ $\sigma_{z,4} = \frac{3}{32}[326.9(3.5) + 326.9(2.5) - 326.9(1.5) - 326.9(0.5)] = 122.59$ Pa.

Summary Table of Interface Stress Amplitudes

Laminate	k	**Location**	$\lvert\tau_{xz}\rvert$ **(Pa)**	$\lvert\sigma_z\rvert$ **(Pa)**	**Primary mechanism**
$[0/90/0/90]_s$	1	0/90	0.00	1.60	Tensile peeling
	2	90/0	0.00	3.21	Tensile peeling
	3	0/90	0.00	4.81	Tensile peeling
	4 (mid)	90/90	0.00	6.41	Peak tensile peeling
$[45/-45/45/-45]_s$	1	$45/-45$	102.53	0.00	Interlaminar shear
	2	$-45/45$	0.00	0.00	Complete shear cancellation
	3	$45/-45$	102.53	0.00	Interlaminar shear
	4 (mid)	$-45/-45$	0.00	0.00	Balanced angle-ply
$[0/90/45/-45]_s$	1	0/90	0.00	0.00	No τ_{xy} or σ_y mismatch
	2	90/45	0.00	31.28	Compressive normal stress
	3	$45/-45$	102.53	78.23	Combined shear and compression
	4 (mid)	$-45/-45$	0.00	94.02	Peak compressive peeling
$[25/-25/90/-90]_s$	1	$25/-25$	174.18	15.32	Interlaminar shear and peel
	2	$-25/90$	0.00	61.29	Tensile peeling
	3	$90/-90$	0.00	107.26	Tensile peeling
	4 (mid)	$-90/-90$	0.00	122.59	Peak tensile peeling

The combined result of V1 and V2 skills result in much more accurate calculations. Despite that, there are still some small issues. First, the shear derivation skips a key justification. Gemini integrates the equilibrium equation and then states that balancing the triangular stress distribution gives the $2/m$ coefficient. It doesn't actually demonstrate the triangular resultant and its moment arm.

Second, the numerical inputs contain some rounding and approximation errors. For laminate C, it uses $\sigma_y = 0$ in the 0° ply. That value is not exactly zero for the stated material properties and loading. Using CLT, the σ_y value for the 0 degree ply should be 1.82 Pa, not zero. This issue was also experienced in the attempt in only using V2 skill, and can be solved by enforcing CLT calculations.

Criterion	**V1 only**	**V2 only**	**V1 + V2**
3D equilibrium	Correct	Correct	Correct
Shear stress mechanism	Partially correct	Correct	Correct
Shear stress amplitude	Incorrect	Correct	Correct
Normal stress mechanism	Partially correct	Correct	Correct
Normal stress amplitude	Incorrect	Correct equation	Correct equation
Numerical implementation	Mostly incorrect	Mixed	Best

MATLAB Code:

```
clear; clc;
%% Material Properties
E1 = 140e9; E2 = 10e9; G12 = 7e9; v12 = 0.3;
v21 = (v12 * E2) / E1;
t_ply = 0.000127;
N = [1;0;0];

%% Ply Sequence Definition
layup = [0, 90, 0, 90, 90, 0, 90, 0];
n_plies = length(layup);
H = n_plies * t_ply;

A = zeros(3,3);

%% A
for i = 1:n_plies
    % Orientation for current ply
    theta = deg2rad(layup(i));
    Q_bar = Q_b(theta, E1, E2, G12, v12, v21);

    % Summation
    A = A + Q_bar * t_ply;
end

%% Global Strain and Ply Stress
% global strain
e_global = A \ N;

% Calculate Stress for every ply first
stress_ply = zeros(3, n_plies);
for i = 1:n_plies
    theta = deg2rad(layup(i));
    Q_bar = Q_b(theta, E1, E2, G12, v12, v21);
    stress_ply(:, i) = Q_bar * e_global;
end

%% Stress Calculation at Interfaces
m = length(layup);
for i = 1: (m-1)
    M_y = 0;
    stress_sum = 0;
    for j = 1:i
        arm = (i - j + 0.5) * t_ply;

        force_y = stress_ply(2, j) * t_ply;

        M_y = M_y + (force_y * arm);

        stress_sum = stress_sum + stress_ply(3,j);
    end

    tauxz(i) = 2/m * stress_sum;
    sigmaz(i) = (M_y * 6) / (m^2*t_ply^2);
end

%% Results
fprintf('Laminate Sequence: [%s]\n', num2str(layup))
```

```
fprintf('Results for Interfaces (1 to %d):\n', m)
fprintf('Interface |       Tau_xz       |       Sigma_z      \n')
for i = 1:m-1
    fprintf('  %2d-%-2d    |  %10.2f Pa    |  %10.2f Pa\n', i, i+1, tauxz(i), sigmaz(
    i))
end

fprintf('A Matrix (MN/m)\n')
disp(A / 1e6);
fprintf('Tau_xz\n')
disp(tauxz)
fprintf('Sigma Z\n')
disp(sigmaz)

%% Q bar function
function Q_b = Q_b(theta, E1, E2, G12, v12, v21)
    denom = 1 - v12 * v21;
    Q11 = E1 / denom;
    Q12 = (v12 * E2) / denom;
    Q21 = (v21 * E1) / denom;
    Q22 = E2 / denom;
    Q66 = G12;

    Q = [Q11, Q12, 0;
         Q21, Q22, 0;
         0,   0,   Q66];
    m = cos(theta);
    n = sin(theta);

    % Transformed Reduced Stiffness Components
    Q11_bar = Q11*m^4 + 2*(Q12 + 2*Q66)*n^2*m^2 + Q22*n^4;
    Q12_bar = (Q11 + Q22 - 4*Q66)*n^2*m^2 + Q12*(n^4 + m^4);
    Q22_bar = Q11*n^4 + 2*(Q12 + 2*Q66)*n^2*m^2 + Q22*m^4;
    Q16_bar = (Q11 - Q12 - 2*Q66)*n*m^3 + (Q12 - Q22 + 2*Q66)*n^3*m;
    Q26_bar = (Q11 - Q12 - 2*Q66)*n^3*m + (Q12 - Q22 + 2*Q66)*n*m^3;
    Q66_bar = (Q11 + Q22 - 2*Q12 - 2*Q66)*n^2*m^2 + Q66*(n^4 + m^4);

    Q_b = [Q11_bar, Q12_bar, Q16_bar;
           Q12_bar, Q22_bar, Q26_bar;
           Q16_bar, Q26_bar, Q66_bar];
end
```

SkillV1:

```
---
name: free-edge-mechanics-v1
description: Derive and numerically estimate free-edge interlaminar shear and peeling stresses from equilibrium, geometry, CLT, and mechanics assumptions without supplied final edge-stress formulas or solver code.
V1      Mechanics-guided independent derivation and calculation
Mission
Given laminate material properties, ply stacking sequence, thickness, and membrane load, independently develop a mechanics-based approximation and calculate numerical tau_xz and sigma_z at every ply interface. Do not stop at qualitative descriptions. Explicitly derive equations from free-body diagrams, stress resultants, centroids, and force/moment equilibrium; show intermediate values and a numerical interface table. If an amplitude is underdetermined, identify exactly which assumption is missing, make a transparent physically motivated approximation where reasonable, and distinguish its prediction from a unique exact solution.
Scope
Uniform axial extension of perfectly bonded, equal-thickness laminate plies with B=0. x is loading direction, y is in-plane transverse direction, z is through thickness. Interface k separates plies k and k+1, numbered from the top. m = total number of plies; h0 = individual ply thickness; H = mh0 is the fixed total laminate thickness at all interfaces. The characteristic free-edge boundary-layer extent is H for every interface, not kh0. State the assumed width convention for b and any wide-plate simplification.
Available mechanics knowledge (not the final solution)
CLT determines far-field in-plane ply stresses: calculate Q, transformed Qbar, A = sum(Qbar*h0), epsilon0 = A\N, and [sigma_x,sigma_y,tau_xy] for each ply. Confirm B=0 from stacking geometry or compute it.
For static equilibrium without body forces: d(sigma_x)/dx + d(tau_xy)/dy + d(tau_xz)/dz = 0; d(tau_xy)/dx + d(sigma_y)/dy + d(tau_yz)/dz = 0; d(tau_xz)/dx + d(tau_yz)/dy + d(sigma_z)/dz = 0.
Uniform extension permits neglecting x-derivatives of the far-field stress components for the simplified analysis.
The y-normal free edge requires sigma_y=tau_xy=tau_yz=0 there; tau_xz and sigma_z on z-normal planes are not automatically zero.
Shear mechanism: in angle plies, opposite tau_xy tendencies must transfer x-directed forces across bonded z-normal interfaces as tau_xz when tau_xy relaxes toward the edge. In ideal 0/90 axial loading tau_xy vanishes; do not fabricate this shear mechanism.
Peeling mechanism: incompatible transverse contraction produces sigma_y. Offset transverse-force resultants and associated tau_yz transfer create a couple, opposed by interfacial sigma_z. Use an isolated k-ply section and account for signed forces, their centroids, and moment arms.
The force and moment balancing method is the same at every interface. For each interface, sum contributions from plies 1..k; never reset the boundary-layer dimension to the isolated-section thickness. For symmetric benchmark configurations, compare top/bottom mirror interfaces and explain any predicted mid-plane peak; do not force numerical symmetry if the derived model violates it.
The full 3D edge solution is difficult. Construct and justify a low-order boundary-layer approximation, for example piecewise linear/triangular stress variations, and determine their resultants and centroid locations using elementary integration. A finite approximation is not an exact pointwise singular-edge stress.
Mandatory solution workflow
Parse inputs and units; identify layup and verify B=0. If necessary request missing material/load data, or clearly label illustrative assumptions.
Calculate each ply's far-field sigma_y and tau_xy using CLT; show Qbar/A/strain and ply stress table.
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For tau_xz: isolate plies 1..k; sketch/describe x-directed forces, traction-free edge, shear transfer on the interface; select a simple edge-relaxation profile. Integrate forces/moments and derive an interface-specific amplitude. Show the derivation, not merely a result.
For sigma_z: isolate plies 1..k; compute each signed transverse force and its moment arm about the interface; draw/describe the opposing peeling-stress couple. Choose and justify a simple profile, integrate its resultant and lever arm, equate moments, and derive an interface-specific amplitude.
Use the same fixed H=m*h0 in every boundary-layer profile. Calculate tau_xz and sigma_z numerically at all m-1 interfaces using your independently derived equations.
Check dimensional consistency, signs, force/moment equilibrium, free-edge conditions, zero shear in cross-ply axial loading, and expected cancellations for angle-ply interfaces. Identify residuals or inconsistencies rather than hiding them.
Return a table with interface number, cumulative tau_xy, cumulative transverse moment, predicted tau_xz, predicted sigma_z, units, and a brief physical interpretation.
Experimental isolation
Do NOT consult, activate, quote, or reuse free-edge-derivations-v2, free-edge-matlab-v3, their reference files, their numerical examples, or any prewritten final free-edge equations. Do not call external numerical solver scripts from other skill versions. You MAY use ordinary arithmetic, algebra, and general-purpose calculation to evaluate your own equations. Do not refuse numerical calculations simply because the exact closed-form reference solution is absent.
Scientific integrity
Equilibrium plus boundary conditions alone does not uniquely determine the complete 3D stress field. State any additional distribution assumptions explicitly. Do not claim independent derivation if a formula was retrieved from another skill. If a chosen approximation is not uniquely justified, label it as a model choice and explain its sensitivity.
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SkillV2: (Revised with additional CLT requirement)

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name: free-edge-derivations-v2
description: Analyze free-edge interlaminar stresses for B=0 laminates using physical force balances and the specific triangular-distribution equations in the supplied AAE590AI project.
Free-edge stresses      V2: mechanics and analytical derivation
Scope
Use the user's force/moment approximation for bonded composite laminates with B=0 under uniform in-plane membrane extension. It is an approximate finite-amplitude free-edge model, not a pointwise exact 3D elasticity solution. Use consistent SI units. Do not use an executable solver or supplied numerical answers in this version.
Physical foundation
Coordinates: x loading, y transverse in-plane/free-edge direction, z thickness.
At a y-normal traction-free edge, tau_xy and sigma_y must relax; tau_xz and sigma_z are interfacial stresses on z-normal planes.
tau_xz: Angle plies develop opposite in-plane tau_xy under axial tension. The relaxation of tau_xy at the edge requires x-directed force transfer across ply interfaces. From x-equilibrium: d(sigma_x)/dx + d(tau_xy)/dy + d(tau_xz)/dz = 0. Uniform extension gives d(tau_xz)/dz = -d(tau_xy)/dy.
sigma_z: Mismatched transverse contraction generates signed sigma_y in bonded plies. Transverse forces, their through-thickness offsets, and tau_yz form couples; the balancing normal traction sigma_z produces peeling at the free edge. Do not attribute tau_xz to a pure 0/90 laminate with tau_xy=0.
Definitions and geometry
m: total ply count; h0: equal thickness of each ply; H=m*h0: total laminate thickness.
h: assumed characteristic edge boundary-layer width; in this project set h=H for all interfaces.
b: in-plane half-width/geometric parameter used in the supplied moment derivation; use the report's wide-plate simplification b >> H when b is unavailable.
k: interface after ply k (1 <= k < m), with plies indexed from the same chosen laminate surface.
Unit width is used in the force/moment derivation. Preserve signed ply stresses and the ply ordering.
Derive the far-field in-plane stresses
Verify B=0 for the supplied layup; do not merely assume symmetry if the sequence is not symmetric.
For each ply calculate transformed reduced stiffness Qbar(theta).
A = sum_j Qbar_j*h0, membrane strain epsilon0 = A^{-1} N (no bending coupling).
Far-field global ply stress [sigma_x,j, sigma_y,j, tau_xy,j]^T = Qbar_j*epsilon0.
Report these stresses before any edge-stress calculation.
Derive tau_xz using the supplied triangular approximation
The project balances moments/resultants of the far-field tau_xy distribution and the triangular edge tau_xz distribution:
M_xy = F_xy*(unit arm)
M_xz = 2F_xz(b-h/3)
F_xy = 2h0(b-h/2)tau_xy^LPT
F_xz = (tau_xz,maxh*unit arm)/2
Equating the couples:
h*(b-h/3)tau_xz,max = 2h0*(b-h/2)*tau_xy^LPT
tau_xz,max = 2*(h0/h)*[(b-h/2)/(b-h/3)]*tau_xy^LPT
For interface k, use the signed cumulative far-field ply shear and h=H=m*h0:
tau_xz,k = (2/m)*[(b-H/2)/(b-H/3)]*sum_{j=1}^k tau_xy,j
When b >> H, the ratio approaches 1:
tau_xz,k = (2/m)*sum_{j=1}^k tau_xy,j
Explain why opposing +/-angle plies can make the cumulative sum zero at selected
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    interfaces. The model's tau_xz amplitude is at the free edge, not a claim that
    tau_xz must vanish there.
Derive sigma_z from transverse-force couples
For each ply j above interface k, the transverse force per unit width is:
F_y,j = sigma_y,j*h0
Its centroid-to-interface moment arm is:
a_j,k = (k-j+0.5)*h0
The signed cumulative transverse-force moment is:
M_y,k = sum_{j=1}^k F_y,j*a_j,k
For the project's assumed opposing triangular normal-traction resultants, use the
    same total-thickness characteristic dimension H=m*h0 at every interface:
F_z = (1/2)sigma_z,max(H/2)*(unit width)
The two equal-and-opposite triangle resultants form a couple with lever arm 2H/3, so
    :
M_z = 2F_z(H/3) = sigma_z,max*H^2/6
Equilibrium of signed moments M_z - M_y,k = 0 gives:
sigma_z,k = 6M_y,k/H^2
= (6/(m^2h0^2))sum_{j=1}^k [sigma_y,jh0*(k-j+0.5)*h0]
Use the sign convention of the project when reporting sigma_z. Explain the peel-
    opening vs compressive meaning of the sign. The force/moment balance and fixed H
    are explicit assumptions of the user's approximation.
Reporting and checks
Show all steps from CLT to far-field stresses to interface sums. Do not ignore small
     stress values by setting them to 0.
Give a table for interfaces 1-2 through (m-1)-m with tau_xz,k and sigma_z,k.
Explain stress sign/cancellation and the free-edge physical mechanisms.
Verify units, cumulative force and moment definitions, and B=0.
For the symmetric cross-ply example in the project, sigma_z peaks at the mid-plane;
    do not silently elevate that example to an independently proved universal claim.
State that the assumed triangles do not resolve local singularity or the full 3D
    field.
Important interpretation
The report contains a sentence saying tau_xz goes to zero at the free edge; its
    figure and derivation instead use a free-edge peak amplitude. Follow the
    explicitly derived traction conditions and flag this wording discrepancy rather
    than silently repeating it.
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References

[1] R. Byron Pipes. Micromechanics prediction of composite properties from constituent properties. Lecture Notes, AAE 555: Mechanics of Composite Materials, 2026.

[2] R. B. Pipes and N. J. Pagano. Interlaminar stresses in composite laminates under uniform axial extension. *Journal of Composite Materials*, 4(4):538–548, 1970. doi: 10.1177/002199837000400409. URL `https://doi.org`.

[3] S. Tian. Creating a skill for an ai agent: Introduction, September 2025. URL `https://cdmhub.org`. Accessed November 8, 2026.

[4] Fachgebiet Leichtbau und Strukturmechanik (LSM). Interlaminare spannungen, 2026. URL `https://www.maschinenbau.tu-darmstadt.de/lsm/forschung_lsm/schwerpunkt_lsm/interlaminare_spannungen/interlaminare_spannungen.de.jsp`. Accessed: 2026-11-07.